

Course ID: SW 302

User Interface (UI) Development

Dr. Mohamed Sami Rakha

Lecture#4

Introduction to CSS

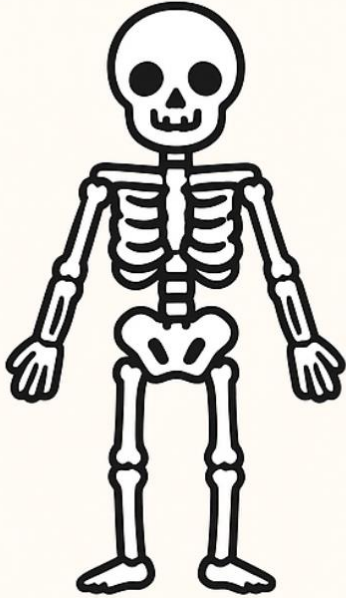
Lecture Outline

Dr. Mohamed Sami Rakha

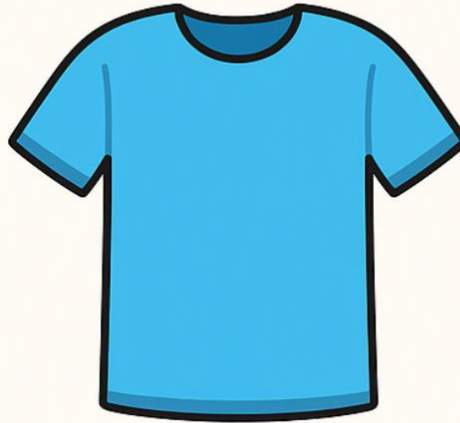
- CSS Inheritance
- CSS Box Model
- CSS Units of Measurement
- Formatting Text
- Colors & Backgrounds
- Floating & Positioning

HTML vs CSS vs JavaScript

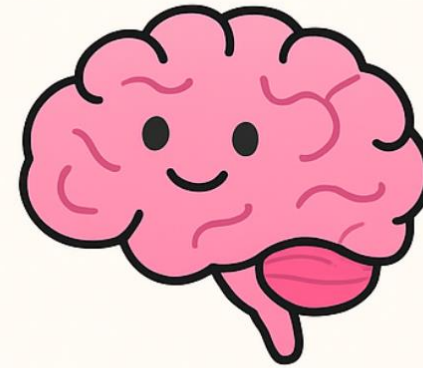
Dr. Mohamed Sami Rakha



HTML
Skeleton
(structure)



CSS
Clothes
(style)



JavaScript
Muscles/Brain
(behavior & interactivity)

The Cascade: Style Weight: Example

```
<p id="note" class="highlight">Hello World</p>
```

p { color: black; } → (Specificity =1)
.highlight { color: green; } → (Specificity =10)
p#note.highlight { color: orange; } → (Specificity =1 (for element)
+ 100 (for ID) + 10 (for class) = 111)
#note { color: red; } → (Specificity =100)

Now, What would be the colour of the text “Hello World”?

The <p> will be **orange**.

Descendent Selectors

A **descendant selector** targets elements contained in another element.

It's a kind of **contextual selector** (it selects based on a relationship to another element).

It's indicated in a list separated by a character space.

```
ol a {font-weight: bold;}
```

(only the links (**a**) in ordered lists (**ol**) would be bold)

```
h1 em {color: red;}
```

(only emphasized text in h1s would be red)

Descendent Selectors

They can appear as part of a grouped selector:

```
h1 em, h2 em, h3 em {color: red;}
```

(only emphasized text in h1, h2, and h3 elements)

They can be several layers deep:

```
ol a em {font-variant: small-caps;}
```

(only emphasized text in links in ordered lists)

Pseudo Class Selectors

Treat elements in a certain state as belonging to the same class

Link Pseudo-classes

:link Applies style to unvisited (unclicked) links

:visited Applies style to visited links

User Action Pseudo-classes

:focus Applies when element is selected for input

:hover Applies when the mouse pointer is over the element

:active Applies when the element (such as a link or button) is in the process of being clicked or tapped

Pseudo Class Selectors

Pseudo Classes must appear in the following order:

```
a { text-decoration: none; } /* turns underlines off for all links */  
a:link { color: maroon; }  
a:visited { color: gray; }  
a:focus { color: maroon; background-color: #ffd9d9; }  
a:hover { color: maroon; background-color: #ffd9d9; }  
a:active { color: red; background-color: #ffd9d9; }
```

Samples of my work:

- Pen and Ink Illustrations
- Paintings
- Collage

`a:link`

Links are maroon and not underlined.

Samples of my work:

- Pen and Ink Illustrations
- Paintings
- Collage

`a:focus`
`a:hover`

While the mouse is over the link or when the link has focus, the pink background color appears.

Samples of my work:

- Pen and Ink Illustrations
- Paintings
- Collage

`a:active`

As the mouse button is being pressed, the link turns bright red.

Samples of my work:

- Pen and Ink Illustrations
- Paintings
- Collage

`a:visited`

After that link has been visited, the link is gray.

Attribute Selectors

Targets elements based on attribute names or values. There

Simple attribute selector

Matches an element with a given attribute:

E[*attribute*]

```
img[title] { border: 3px solid; }
```

(Matches every `img` element that has a `title` attribute)

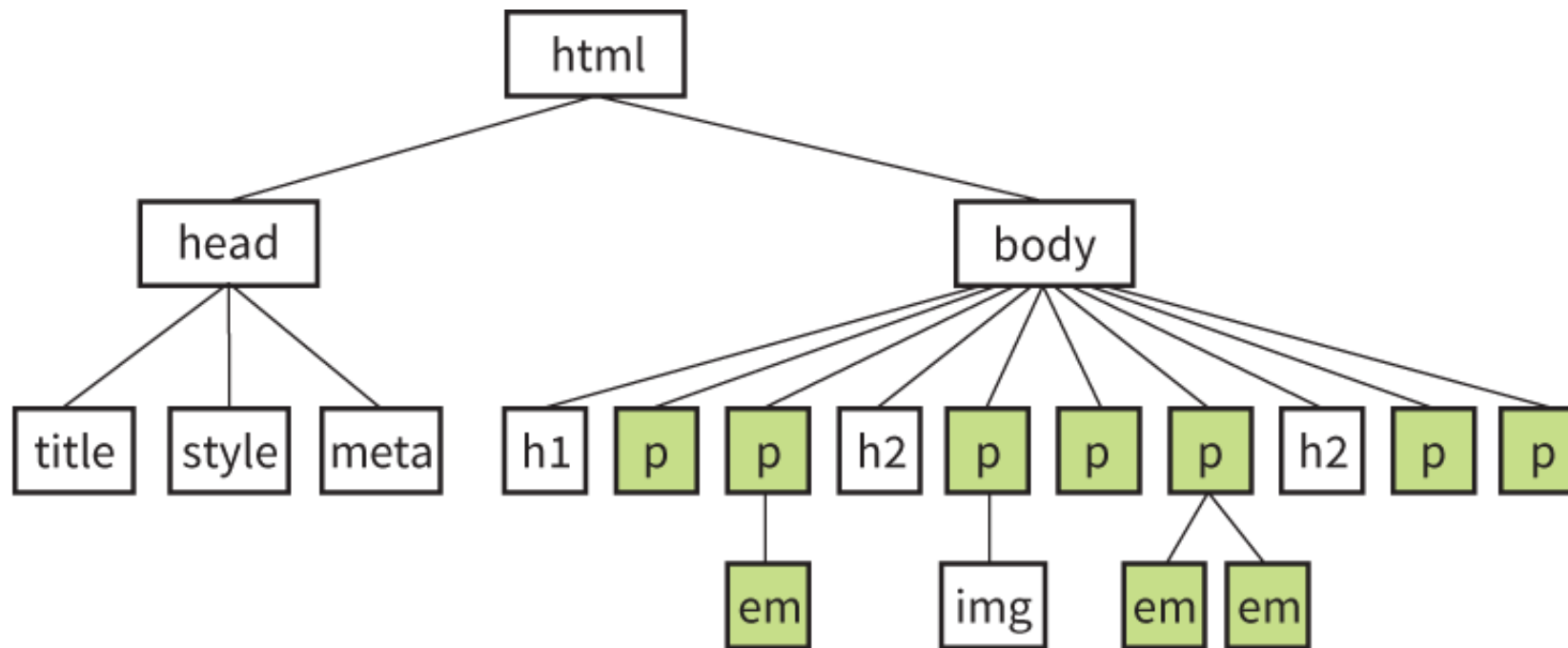
```
img[title="first grade"] {border: 3px solid;}
```

(matches only if the `title` value is "first grade")

CSS Inheritance

CSS Inheritance

- **Many properties** applied to elements are passed down to the elements they contain. This is called inheritance
- For example, applying “**a sans serif**” font to a **<p>** element causes the **** element it contains to be sans serif as well:



`p {font-size: large; font-family: sans-serif;}`

CSS Inheritance

- What if no style declaration applies to a **property of an element**?
- Generally, the property value is inherited **from the nearest ancestor element** that has a value for the property
- If no ancestor has a value (or the property does not inherit) then CSS defines an initial value that is used

```
<body>
  <ul>
    <li>
      List item outside and <span>inside</span> a span.
      <p>
        Embedded paragraph outside and <span>inside</span> a span.
      </p>
    </li>
  </ul>
</body>
```

```
body { font-weight: bold}
li { font-style: italic}
p { font-size: larger}
span { font-weight: normal}
```

- *List item outside and inside a span.*

Embedded paragraph outside and inside a span.

CSS Inheritance

- Nested elements inherit the properties from the its parent.
- If you specify a style for the `<body>` tag it will affect all content in your HTML page.
- If you want to override inherited settings, you need to specify a style in a more local element



```
body {  
    font-family: Verdana;  
    font-size: 14px;  
}
```

```
body {          font-family: Verdana;  
    font-size: 1.1em; }  
.littletext { font-size: 8px; }  
  
<body>  
This text is larger.  
<p class="littletext">This text is  
smaller.</p>
```

CSS Inheritance

- Some properties inherit; others do not. Properties related to text usually inherit; properties related to layout generally don't.
- Styles explicitly applied to specific elements **override inherited** styles.
- You'll learn to use inheritance strategically to keep your style rules simple.

CSS Box Model

CSS Box Model

- **Browsers** see every element on the page as being contained in a little rectangular box. **Block elements and inline elements participate in the box model.**
- In this example, a blue border is added to all elements.

Cooking with Daniel from Nada Surf

I had the pleasure of spending a crisp, Spring day in Portsmouth, NH cooking and chatting with Daniel Lorca of the band Nada Surf as he prepared a gourmet, sit-down dinner for 28 pals.

When I first invited Nada Surf to be on the show, I was told that Daniel Lorca was the guy I wanted to talk to. The response: "I'm way into it, but I don't want to talk about it, I wanna do it." After years of only having access to their sound check and set, I've been doing a lot of *talking* about cooking with rockstars. To actually cook is true.

Six-hour Salad



Daniel prepared a salad of arugula, smoked tomatoes, tomato jam, and grilled avocado (it's as good as it sounds!). I jokingly called it "6-hour Salad" because that's how long he worked on it. The fresh tomatoes were slowly smoked over woodshims in the grill, and when

I was told
talk about it, I
of *talking* at

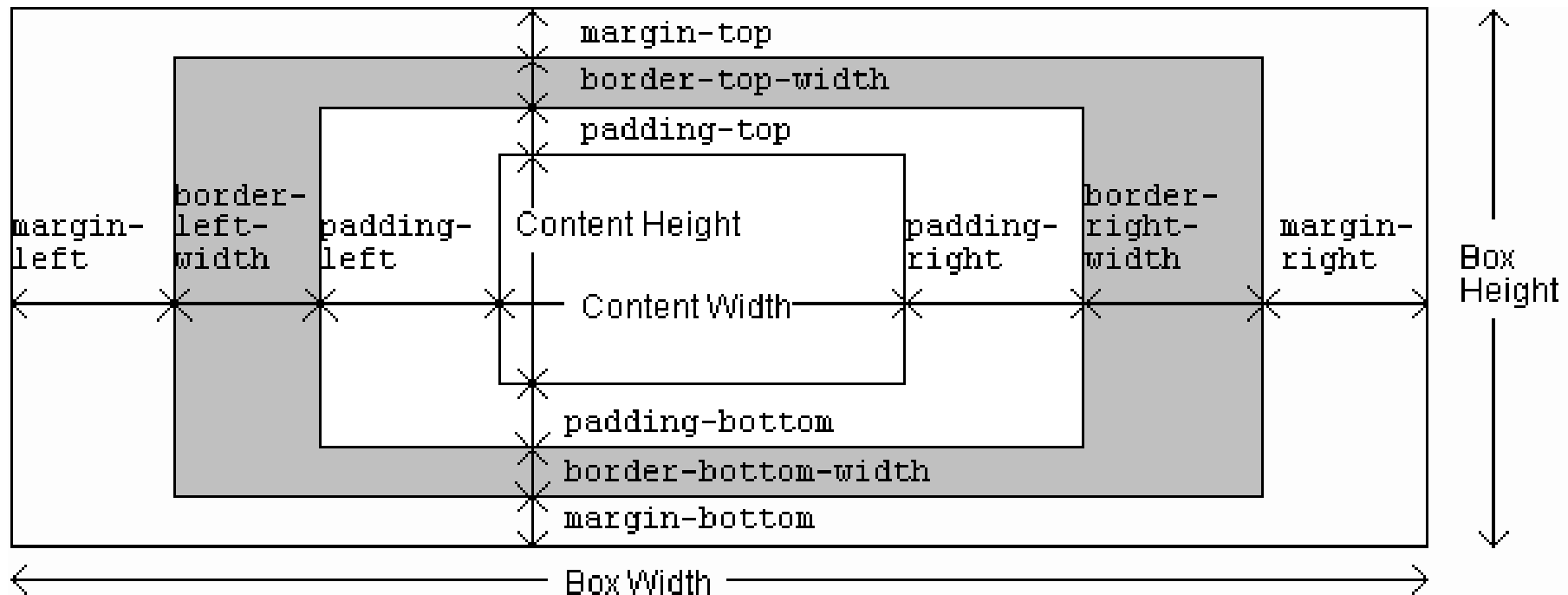
CSS Box Model

Dr. Mohamed Sami Rakha

- The **box model** is the **foundation of CSS page layout.**
- Apply properties such as **borders, margins, padding, and backgrounds** to element boxes.
- Position, move, grow, and shrink boxes to create fixed or flexible page layouts.

CSS Box Model

- When the browser draws an object on a page, it places it into an invisible rectangular space called a “bounding box.”
- You can specify the size, look, and feel of the margins, the padding, the border, and the content of the box.



CSS Box Model

Basic CSS style properties associated with the box model.

ni Rakha

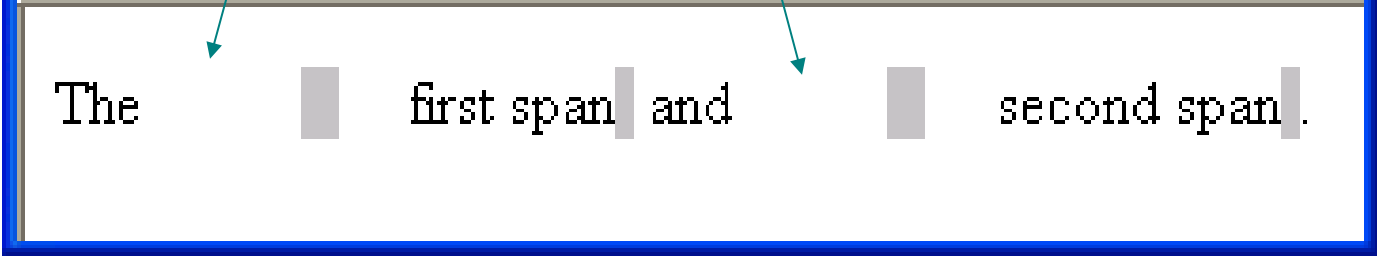
Property	Values
padding-{top,right,bottom,left}	CSS length (Sec. 3.6.2).
padding	One to four length values (see text).
border-{top,right,bottom,left}	One to three values (in any order) for border-*-width, border-*-color, and border-*-style. Initial values are used for any unspecified values.
border	One to three values; equivalent to specifying given values for each of border-top, border-right, border-bottom, and border-left.
margin-{top,right,bottom,left}	auto (see text) or length.
margin	One to four margin-* values.
border-{top,right,bottom,left}-width	thin, medium (initial value), thick, or a length.
border-width	One to four border-*-width values.
border-{top,right,bottom,left}-color	Color value. Initial value is value of element's color property.
border-color	transparent or one to four border-*-color values.

CSS Box Model – Margin Example

Dr. Mohamed Sami Rakha

```
span { margin-left: 1cm;  
        border-left-width: 10px;  
        border-left-color: silver;  
        border-left-style: solid;  
        padding-left: 0.5cm;  
        border-right-width: 5px;  
        border-right-color: silver;  
        border-right-style: solid }
```

Note that there are two tags used (first and second) in the example below

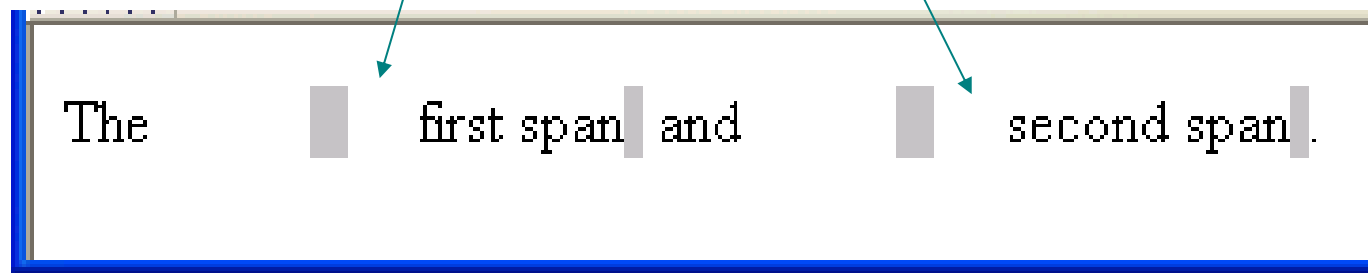


The first span and second span.

CSS Box Model – Padding Example

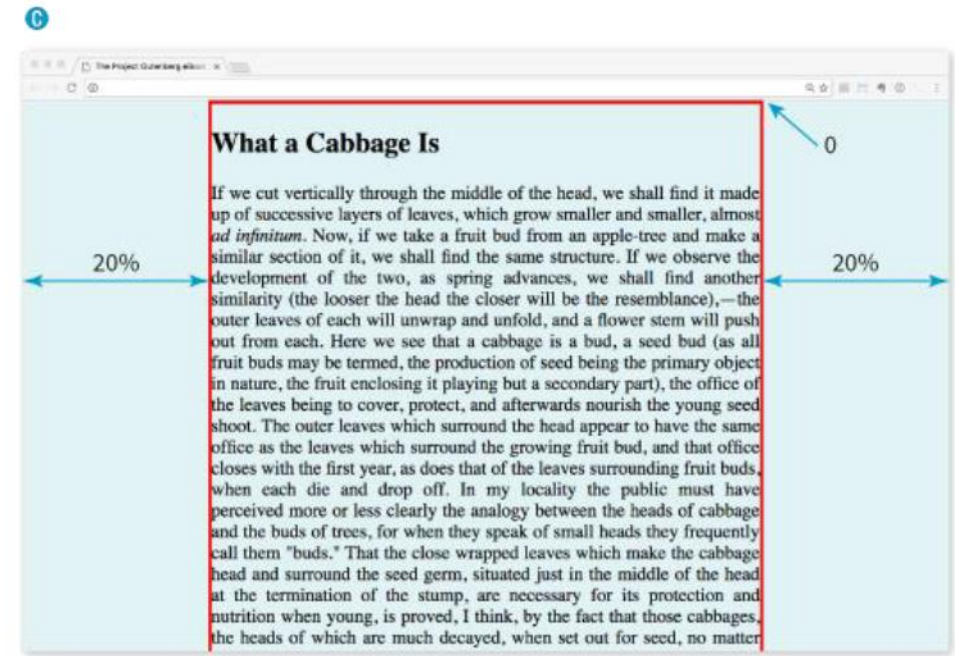
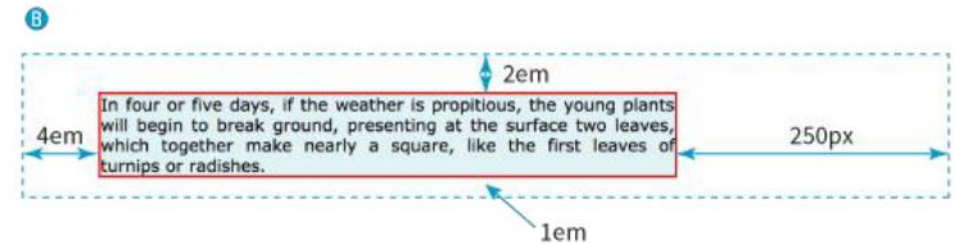
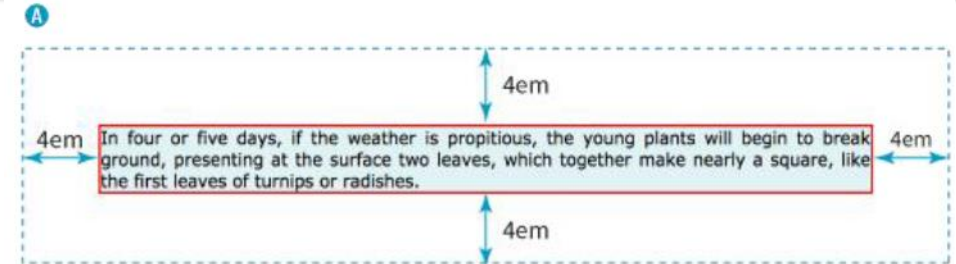
```
span { margin-left: 1cm;  
        border-left-width: 10px;  
        border-left-color: silver;  
        border-left-style: solid;  
        padding-left: 0.5cm;  
        border-right-width: 5px;  
        border-right-color: silver;  
        border-right-style: solid }
```

Note that there are two tags used (first and second) in the example below



CSS Box Model – Another Margin Example

- A** `p#A {`
 `margin: 4em;`
 `border: 2px solid red;`
 `background: #e2f3f5;`
}
- B** `p#B {`
 `margin-top: 2em;`
 `margin-right: 250px;`
 `margin-bottom: 1em;`
 `margin-left: 4em;`
 `border: 2px solid red;`
 `background: #e2f3f5;`
}
- C** `body {`
 `margin: 0 20%;`
 `border: 3px solid red;`
 `background-color: #e2f3f5;`
}



Borders

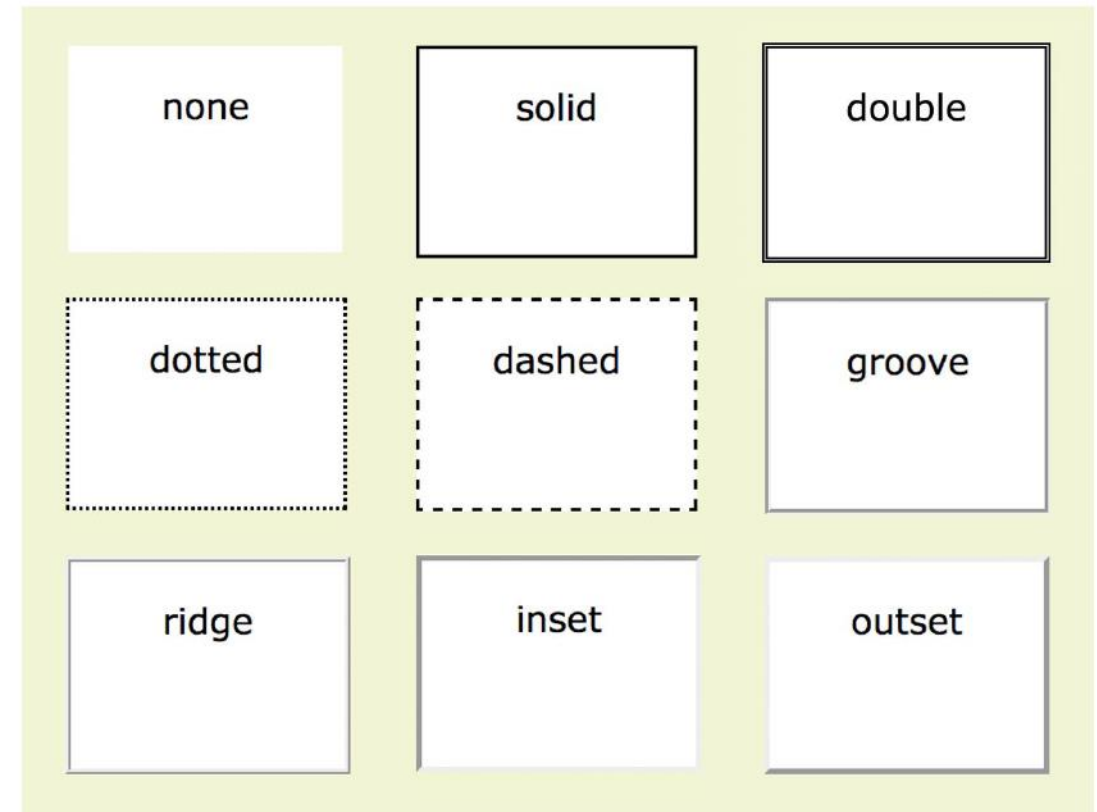
- A **border** is a line drawn around the content area and its (optional) padding.
- The thickness of the border is included in the dimensions of the element box.
- You define **style**, **width**(thickness), and **color** for borders.
- Borders can be applied to single sides or all around

Border Style

border-style, border-top-style, border-right-style, border-bottom-style, border-left-style

Values: none, solid, hidden, dotted, dashed, double, groove, ridge, inset, outset

NOTE: The default is none, so if you don't define a border style, it won't appear.

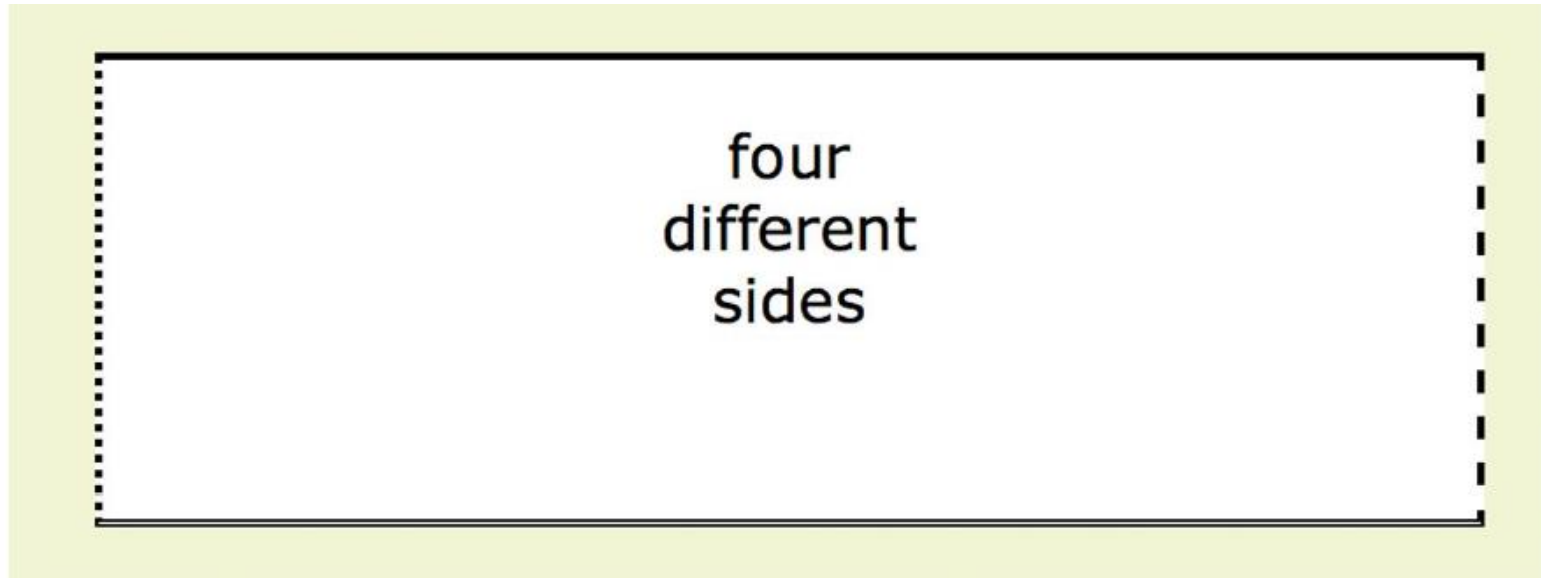


Border Style

The **border-style** shorthand uses the clockwise shorthand order. The following rules have the same effect:

```
div#silly {  
  border-top-style: solid;  
  border-right-style: dashed;  
  border-bottom-style: double;  
  border-left-style: dotted;  
  width: 300px;  
  height: 100px;  
}
```

```
div#silly {  
  border-style: solid dashed double dotted;  
}
```

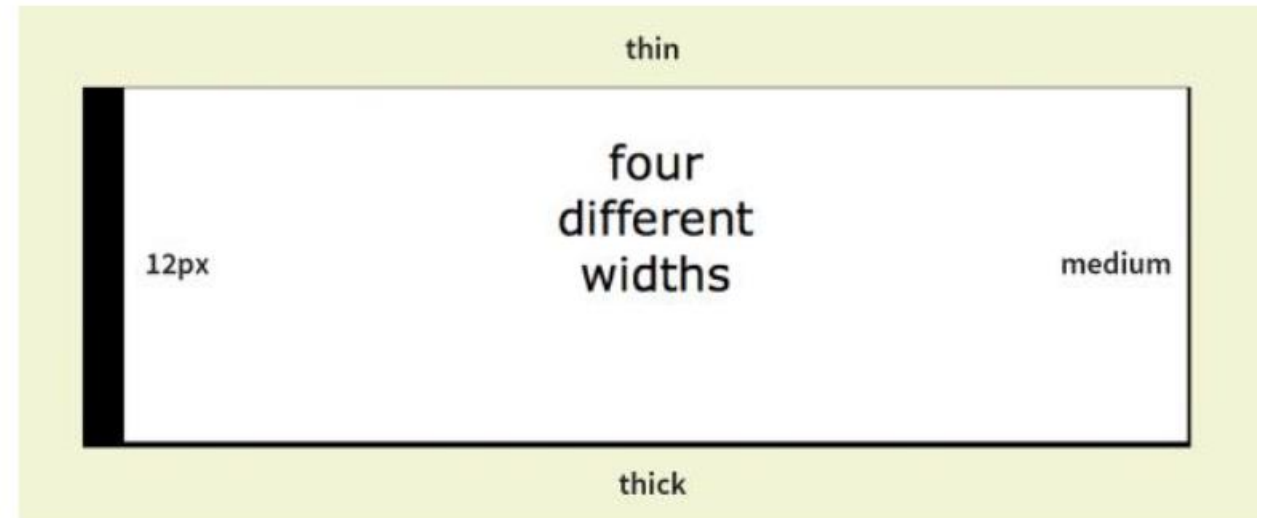


Border Width

border-width, border-top-width, border-right-width, border-bottom-width, border-left-width, Values: Length, thin, medium, thick

```
div#help {  
  border-width: thin medium thick 12px;  
  border-style: solid;  
  width: 300px;  
  height: 100px;  
}
```

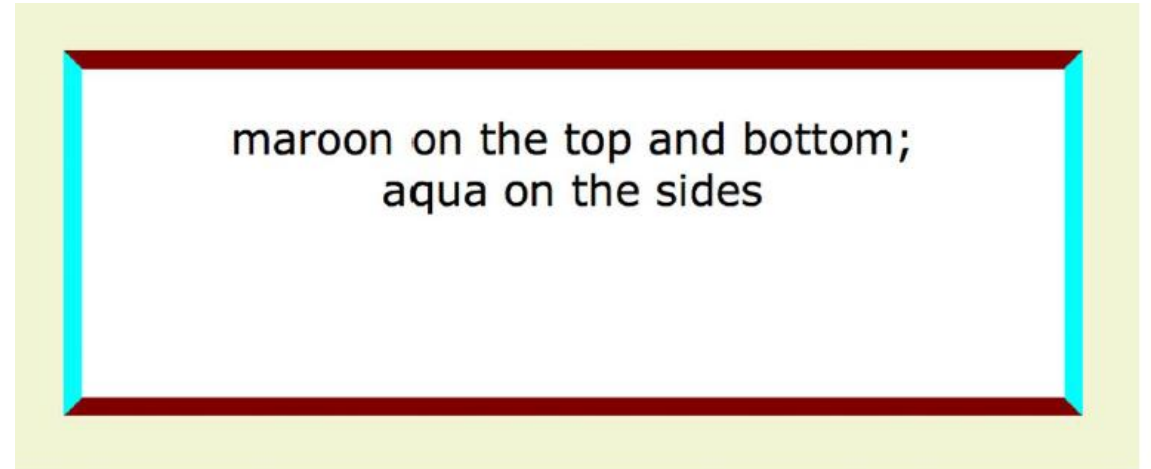
NOTE: The `border-style` property is required for the border to be rendered.



Border Color

Border color , border top color , border right color border bottom color , border left color, **Values:** Color value (named or numeric)

```
div#special {  
    border-color: maroon aqua;  
    border-style: solid;  
    border-width: 6px;  
    width: 300px;  
    height: 100px;  
}
```



NOTE: The `border-style` property is required for the border to be rendered.

Border Shorthand Properties

Values: border-style border-width border-color
Combine style, width, and color values in shorthand properties for each side or all around (**border**):

```
p.example {  
    border: 2px dotted aqua;  
}
```

NOTE: The **border-style** property must be included in the shorthand for the border to be rendered.

Border Radius (Rounded Corners)

Border radius, Values: 1, 2, 3, or 4 length or percentage values

- The **border-radius** property rounds off the corners of an element.
- The value is a length or percentage value reflecting the radius of the curve.
- Providing one value makes all the corners the same.
- Four values are applied clockwise, starting from the top-left corner.

Border Radius (Rounded Corners)

```
p {  
  width: 200px;  
  height: 100px;  
  background: darkorange;  
}
```



```
border-radius: 1em;
```



```
border-top-right-radius: 50px;
```



```
border-radius: 50px;
```



```
border-top-left-radius: 1em;  
border-top-right-radius: 2em;  
border-bottom-right-radius: 1em;  
border-bottom-left: 2em;
```

or

```
border-radius: 1em 2em;
```

mi Rakha

CSS Units of Measurement

CSS Units of Measurement

CSS provides a variety ways to specify measurements:

Absolute units

Have predefined meanings or real world equivalents

Relative units

Based on the size of something else, such as the default text size or the size of the parent element

Percentages

Calculated relative to another value, such as the size of the parent element

Absolute Units

With the exception of pixels, absolute units are not appropriate for web design? Why ?

px	pixel
in	inches
mm	millimeters
cm	centimeters
q	1/4 millimeter
pt	points (1/72 inch)
pc	pica (1 pica = 12 points = 1/6 inch)

Absolute Units

Dr. Mohamed Sami Rakha

With the exception of pixels, **absolute units are not appropriate for web design**? Why ?

Answer: Absolute units are *fixed physical measurements*. They don't scale based on screen size, zoom level, or device pixel density.

Relative Units

Relative units are based on the size of something else:

em	a unit equal to the current font size
ex	x-height, equal to the height of a lowercase <i>x</i>
rem	root em, equal to the font size of the <code>html</code> element
ch	zero width, equal to the width of a zero (0)
vw	viewport width unit (equal to 1/100 of viewport width)
vh	viewport height unit (1/100 of viewport height)
vmin	viewport minimum unit (value of <code>vh</code> or <code>vw</code> , whichever is smaller)
vmax	viewport maximum unit (value of <code>vh</code> or <code>vw</code> , whichever is larger)

Relative Units

Dr. Mohamed Sami Rakha

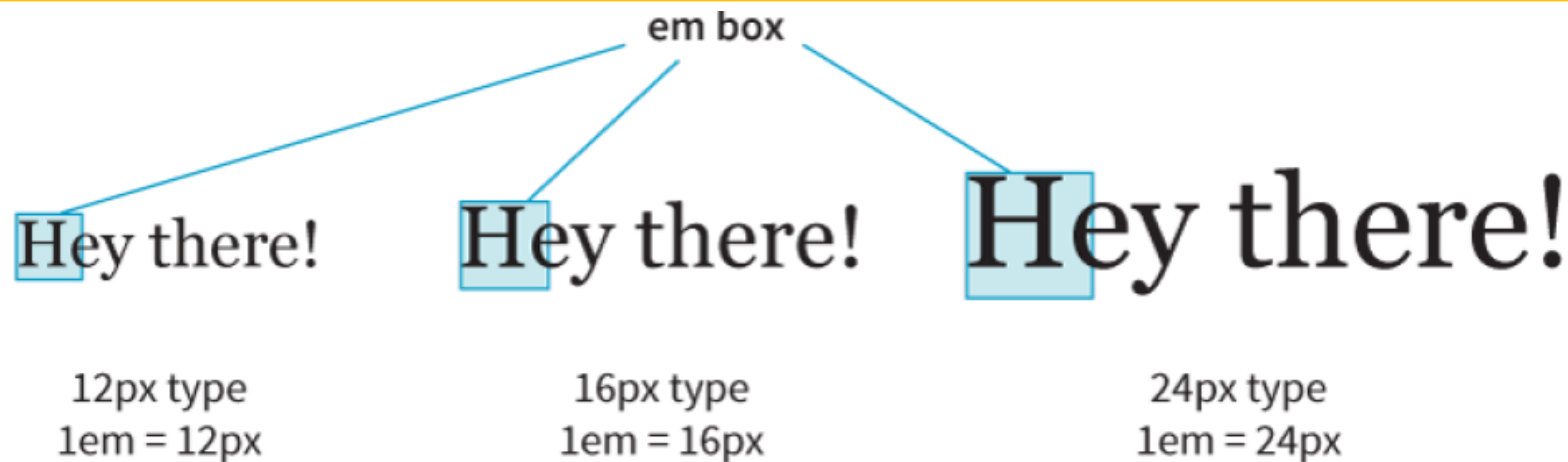
Relative units scale depending on the user's settings or the parent element's size, **making them flexible and responsive.**

The rem Unit

- The **rem(root em)** unit is based on the font size of the html (<html> tag) element, whatever that happens to be.
- **Default** in modern browsers: Root font size is 16 pixels, so a **rem = a 16-pixel unit**.
- If the root font size of the document changes, so does the size of a rem (and that's good for keeping elements proportional).

The em Unit

- The em unit is traditionally based on the width of a capital letter M in the font. measures **size relative to the font-size of the element it's used in (or its parent)**. When the font size is 16 pixels, then **1 em = 16 pixels, 2 em = 32 pixels, and so on.**



NOTE: Because they're based on the font size of the current element, the size of an em may not be consistent across a page.

The em Unit: example

```
body {  
  font-size: 16px;  
}  
  
p {  
  font-size: 1.5em;  
}
```

Dr. Mohamed Sami Rakha
Because it inherits the parent
(body) font size of 16px.

Each `<p>` element will have
 $1.5 \times 16\text{px} = 24\text{px}$ font size.

The em Unit: Nested em values example

```
div {  
  font-size: 20px;  
}  
  
p {  
  font-size: 1.2em; /* relative to div  
  */  
}  
  
span {  
  font-size: 1.2em; /* relative to p  
  */  
}
```

HTML: <div><p> Hello!
</p></div>

What is the Font size of “Hello!”?

Element	Calculation	Resulting font size
div	base	20px
p	$1.2 \times 20\text{px}$	24px
span	$1.2 \times 24\text{px}$	28.8px

Viewport Percentage Lengths (vw/vh)

Viewport width (**vw**) and viewport height (**vh**) units are relative to the size of the **viewport (browser window, meaning screensize!)**:

vw = 1/100th width of viewport

vh = 1/100th height of viewport

They're useful for making an element fill the viewport or a specified percentage of it. This image will be 50% the width and height of the viewport:

```
img { width: 50vw; height: 50vh; }
```

Formatting Text

Designing Text

- Styling text on the web is tricky because you don't have control over how the text displays for the user:
- They may not have the font you specify.
 - They may have their text set larger or smaller than you designed it.
- **Best practices allow for flexibility in text specification.**

CSS properties for fonts



CSS font properties deal with specifying the shapes of the characters themselves:

property	description
font-family	which font will be used
font-size	how large the letters will be drawn
font-style	used to enable/disable italic style
font-weight	used to enable/disable bold style

- [Complete list of font properties](https://www.w3schools.com/css/css_font.asp)

https://www.w3schools.com/css/css_font.asp

Specifying the Font Family

font-family property defines the typeface used for text.

- Font names must be capitalized (except generic font families).
- Use commas to separate multiple font names.
- If the name has a character space, it must appear within quotation marks:

```
body { font-family: Arial; }  
var { font-family: Courier, monospace; }  
  
p { font-family: "Duru Sans", Verdana, sans-serif; }
```

- **Enclose multi-word font names in quotes**

Specifying the Font Family: Example

```
p {  
  font-family: Georgia;  
}  
h2 {  
  font-family: "Courier New";  
}
```

kha

CSS

Output:

This paragraph uses the first style above.

This h2 uses the second style above.

More about Font-Family

Dr. Mohamed Sami Rakha

```
p {  
font-family: Garamond, "Times New Roman", serif;  
}
```

CSS

Output: This paragraph uses the above style.

- We can specify multiple fonts from highest to lowest priority
- Generic font names: serif, sans-serif, cursive, FANTASY, monospace
- If the first font is not found on the user's computer, the next is tried
- Placing a generic font name at the end of your font-family value, ensures that every computer will use a valid font

More about Font-Family

The font must be available on the user's machine for it to display.

- The best practice is to provide a list of options. The browser uses the first one that is available. "Generic font families are often used as the last backup option."

Start with the font you want and then provide backup options ending with a generic font family, as shown here:

```
p { font-family: "Duru Sans", Verdana, sans-serif; }
```

You can also download a web font with the page, but it adds to the download and display time. **Not all Font Families are FREE!**

Generic Font Families

Generic font families instruct the browser to use an available font from one of five stylistic categories: **serif**, **sans-serif**, **monospace**, **cursive**, **fantasy**

Generic font families are often used as the last backup option.

serif



Hello
Times

Hello
Georgia

Hello
Times New Roman

Hello
Lucida

sans-serif



Hello
Verdana

Hello
Trebuchet MS

Hello
Arial

Hello
Arial Black

monospace

W i

Monospace font
(equal widths)

W i

Proportional font
(different widths)

Hello
Courier

Hello
Courier New

Hello
Andale Mono

cursive

Hello
Apple Chancery

Hello
Comic Sans

Hello
Snell

fantasy

Hello
Impact

HELLO
Stencil

HELLO
Mojo

Specifying Font Size

Dr. Mohamed Sami Rakha

Values:

- CSS length units
- Percentage value
- Absolute keywords (xx-small, x-small, small, medium, large, x-large, xx-large)
- Relative keywords (larger, smaller)

Example:

```
h1 { font-size: 2.5rem; }
```

Specifying Font Size

Most common sizing methods:

- **rem** and **em** units.
- **percentages**(based on the inherited font size for that element).
- **pixels(px)** can be used, but they're not flexible.

Absolute-size keywords

xx-small | x-small | small | medium | large | x-large | xx-large : They are called absolute keywords because they define specific relative steps in size, not “absolute” in pixels, **but fixed in relation to each other.** Let’s say the browser’s default medium size is 16px (which it usually is).

Dr. Mohamed Sami Rakha

Keyword	Typical computed size (if <code>medium</code> = 16px)
<code>xx-small</code>	9px
<code>x-small</code>	10px
<code>small</code>	13px
<code>medium</code>	16px
<code>large</code>	18px
<code>x-large</code>	24px
<code>xx-large</code>	32px

Font-Size

```
p {  
    font-size: 24pt;  
}
```

CSS

Output:

This paragraph uses the style above.

- units: pixels (px) vs. point (pt) vs. m-size (em)

16px, 16pt, 1.16em

- vague font sizes: xx-small, x-small, small, medium, large, x-large, xx-large, smaller, larger
- percentage font sizes, e.g.: 90%, 120%

Font Weight (Boldness)

Values: normal, bold, bolder, lighter, 100, 200, 300, 400, 500, 600, 700, 800, 900

Example:

```
h1 { font-weight: normal; }
```

```
span.new { font-weight: bold; }
```

- Most common values are normal and bold.
- Numerical values are useful when using a font with multiple weights.

Font Style (Italics)

Values: normal, italic, oblique

Dr. Mohamed Sami Rakha

Example:

```
cite { font-style: italic; }
```

- Makes text italic, normal, or oblique (slanted, but generally the same as italics).

Small Caps

Values: normal , small caps

Dr. Mohamed Sami Rakha

Example:

```
abbr { font-variant: small-caps; }
```

- Small caps are a separate font design that uses small uppercase characters in place of lowercase letters.
- They help acronyms and other strings of capital letters blend in with the weight of the surrounding text.

Condensed and Extended Text

Values: normal, ultra-condensed, extra-condensed, condensed, semi-condensed, semi-expanded, expanded, extra-expanded, ultra-expanded

Example:

```
abbr { font-variant: small-caps; }
```

- Tells the browser to select a normal, condensed, or extended font variation from a typeface if it is available

Design

Universe Ultra Condensed

Design

Universe Condensed

Design

Universe

Design

Universe Extended

Some properties control whole lines of text:

- Line height (line-height)
- Indents (text-indent)
- Horizontal alignment (text-align)

Line Height

Dr. Mohamed Sami Rakha

Values: Number, length, percentage, normal

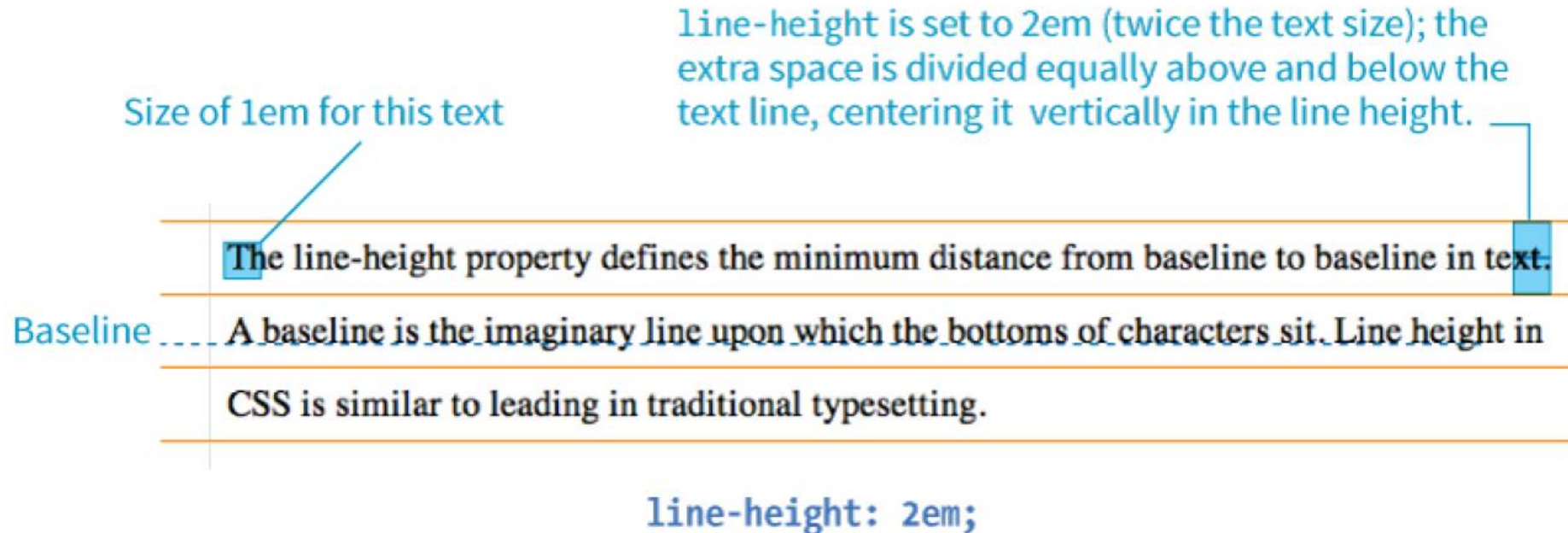
Example:

```
p { line-height: 1.4em; }
```

Line height defines the minimum distance from baseline to baseline in text.

Line Height

- The **baseline** is the imaginary line upon which the bottoms of characters sit.
- If a large character or image is on a line, the line height expands to accommodate it.



Indents

Values: Length, percentage

Dr. Mohamed Sami Rakha

Examples:

```
p {text-indent: 2em; }
```

```
p {text-indent: 25%; }
```

```
p {text-indent: -35px; }
```

 Paragraph 1. The text-indent property indents only the first line of text by a specified amount. You can specify a length measurement or a percentage value.

 Paragraph 2. The text-indent property indents only the first line of text by a specified amount. You can specify a length measurement or a percentage value.

 Paragraph 3. The text-indent property indents only the first line of text by a specified amount. You can specify a length measurement or a percentage value.

Text-Align

Values: left,
right, center,
justify, start,
end

```
text-align: left;
```

Paragraph 1. The text-align property controls the horizontal alignment of the text within an element. It does not affect the alignment of the element on the page. The resulting text behavior of the various values should be fairly intuitive.

```
text-align: right;
```

Paragraph 2. The text-align property controls the horizontal alignment of the text within an element. It does not affect the alignment of the element on the page. The resulting text behavior of the various values should be fairly intuitive.

```
text-align: center;
```

Paragraph 3. The text-align property controls the horizontal alignment of the text within an element. It does not affect the alignment of the element on the page. The resulting text behavior of the various values should be fairly intuitive.

```
text-align: justify;
```

Paragraph 4. The text-align property controls the horizontal alignment of the text within an element. It does not affect the alignment of the element on the page. The resulting text behavior of the various values should be fairly intuitive.

Underlines (Text Decoration)

Values: none, underline, overline, line-through, blink

Examples: I've got laser eyes.

```
text-decoration: underline;
```

I've got laser eyes.

```
text-decoration: overline;
```

~~I've got laser eyes.~~

```
text-decoration: line-through;
```

Underlines (Text Decoration)

by default, links in HTML are underlined. Sami Rakha

- If you turn off underlines under links, be sure there is another visual cue to compensate.
- Underlining text that is not a link may be misleading. Consider italics instead.

NOTE:

`text-decoration` is often used to turn **off** underlines under links:

```
a {  
    text-decoration: none;  
}
```


CSS Properties for Text

property	description
text-align	alignment of text within its element
text-decoration	decorations such as underlining
line-height, word-spacing, letter-spacing	gaps between the various portions of the text
text-indent	indents the first letter of each paragraph

https://www.w3schools.com/css/css_text_align.asp

Colors & Backgrounds

CSS Properties for Colors

```
p {  
  color: red;  
  background-color: yellow;  
}
```

CSS

Output:

This paragraph uses the style above

property	description
color	color of the element's text
background-color	color that will appear behind the element

CSS Properties for Colors

- Specify foreground or background color using one of 140 predefined **color names**:

```
h1 { color: red; }
```

```
h2 { color: darkviolet; }
```

```
body { background-color: papayawhip; }
```

Specifying colors

```
p { color: red; }  
h2 { color: rgb(128, 0, 196); }  
h4 { color: #FF8800; }
```

CSS

Output:

This paragraph uses the first style above

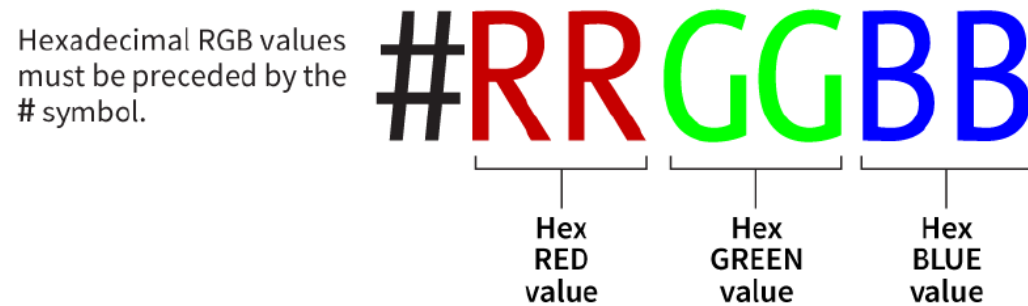
This h2 uses the second style above.

This h4 uses the third style above.

- **Color names:** aqua, black, blue, fuchsia, gray, green, lime, maroon, navy, olive, purple, red, silver, teal, white (white), yellow
- **RGB codes:** red, green, and blue values from 0 (none) to 255 (full), **RGBA**(RGB plus alpha transparency)
- **Hex codes:** RGB values in base-16 from 00 (0, none) to FF (255, full)

Hexadecimal RGB Values

Red, green, and blue values converted to **hexadecimal** and preceded by the **#**symbol.



Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F



The decimal number **32** is represented as

20
2 sixteens and 0 ones

The decimal number **42** is represented as

2A
2 sixteens and 10 ones

RGBA Color

RGB + an alpha channel for transparency

Dr. Mohamed Sami Rakha

The first three values are RGB. The fourth is the transparency level from 0 (transparent) to 1 (opaque).

Playing with RGBA

Playing with RGBA

Playing with RGBA

```
color: rgba(0, 0, 0, .1);
```

```
color: rgba(0, 0, 0, .5);
```

```
color: rgba(0, 0, 0, 1);
```

Grouping styles

```
p, h1, h2 {  
  color: green;  
}  
h2 {  
  background-color: yellow;  
}
```

CSS

Output:

This paragraph uses the above style.

This h2 uses the above styles.

- A style can select multiple elements separated by commas
- The individual elements can also have their own styles

Foreground Color

- **Values:** Color value(named or numeric)

Example: `blockquote {border: 4px dashed; color: green; }`

The **foreground** of an element consists of its text and border (if one is specified).

In the latitude of central New England, cabbages are not secure from injury from frost with less than a foot of earth thrown over the heads. In mild winters a covering of half that depth will be sufficient; but as we have no prophets to foretell our mild winters, a foot of earth is safer than six inches.

Background Color

- **Values:** Color value(named or numeric)

Example: `blockquote {border: 4px dashed;
background-color: green;}`

- The **background painting area** of an element fills the area behind the text to the outer edge of the border.

In the latitude of central New England, cabbages are not secure from injury from frost with less than a foot of earth thrown over the heads. In mild winters a covering of half that depth will be sufficient; but as we have no prophets to foretell our mild winters, a foot of earth is safer than six inches.

Clipping the Background

- **Values:** border-box, padding-box, content-box
- **Specifies where the background painting area ends.**

In the latitude of central New England,
cabbages are not secure from injury from
frost with less than a foot of earth thrown
over the heads.

`background-clip: border-box;`

In the latitude of central New England,
cabbages are not secure from injury from
frost with less than a foot of earth thrown
over the heads.

`background-clip: padding-box;`

In the latitude of central New England,
cabbages are not secure from injury from
frost with less than a foot of earth thrown
over the heads.

`background-clip: content-box;`

Opacity

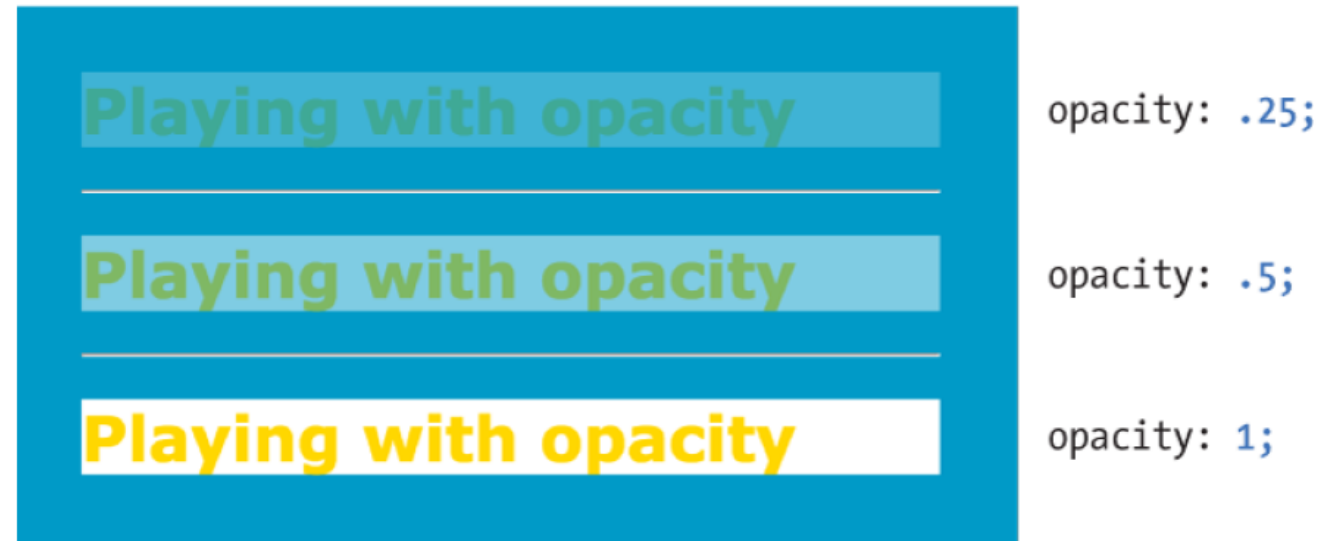
- **Values:** number(0 to 1)

Dr. Mohamed Sami Rakha

- Specifies the transparency level from 0 (transparent) to 1 (opaque)

Example:

```
h1 {color: gold; background: white; opacity: .25;}
```



Background-Image

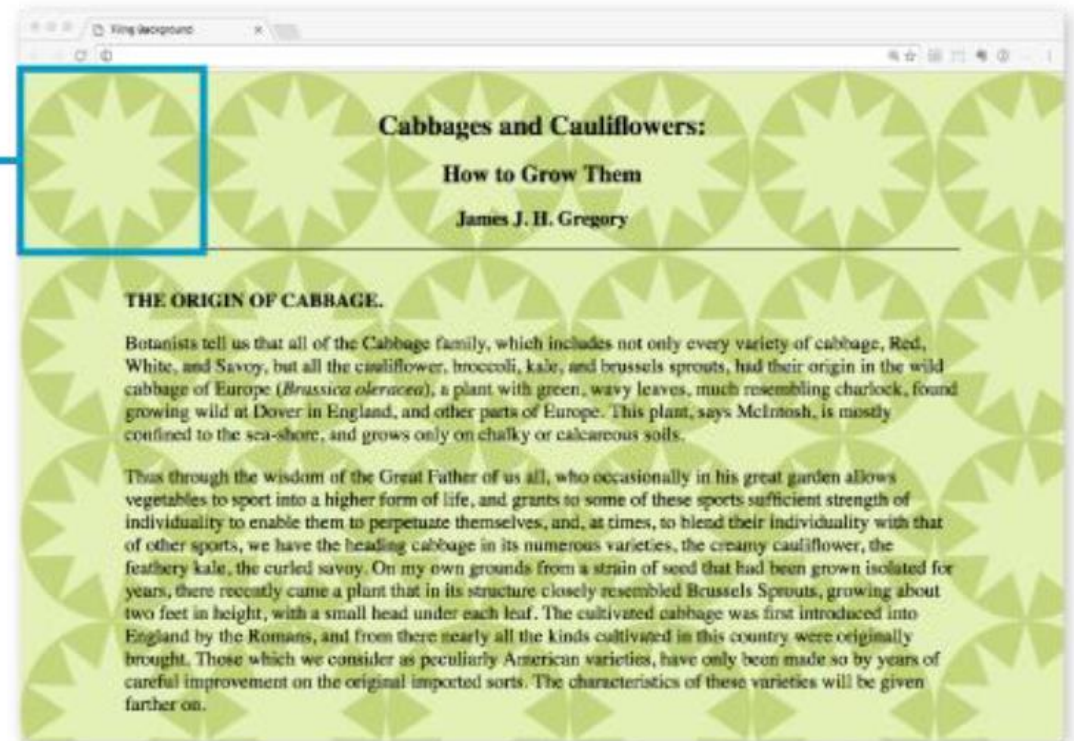
- **Values:** number(0 to 1)

```
body {  
  background-image: url("images/star.jpg"); }  
CSS
```

- Locates an image to be used as a tiling background image behind an element. By default, it starts at the top, left corner and repeats horizontally and vertically:



star.png
150 × 150 pixels



Background-Repeat

- **Values:** repeat, no-repeat, repeat-x, repeat-y, space, round
- Specifies how the background image repeats and can restrict it to tiling in one direction or not at all:
 - **repeat-x:** Tiles horizontally only
 - **repeat-y:** Tiles vertically only
 - **space:** Adds space around images so they fit in the window with no clipping
 - **round:** Distorts the image so it fits without clipping
- Can be repeat (default), repeat-x, repeat-y, or no-repeat

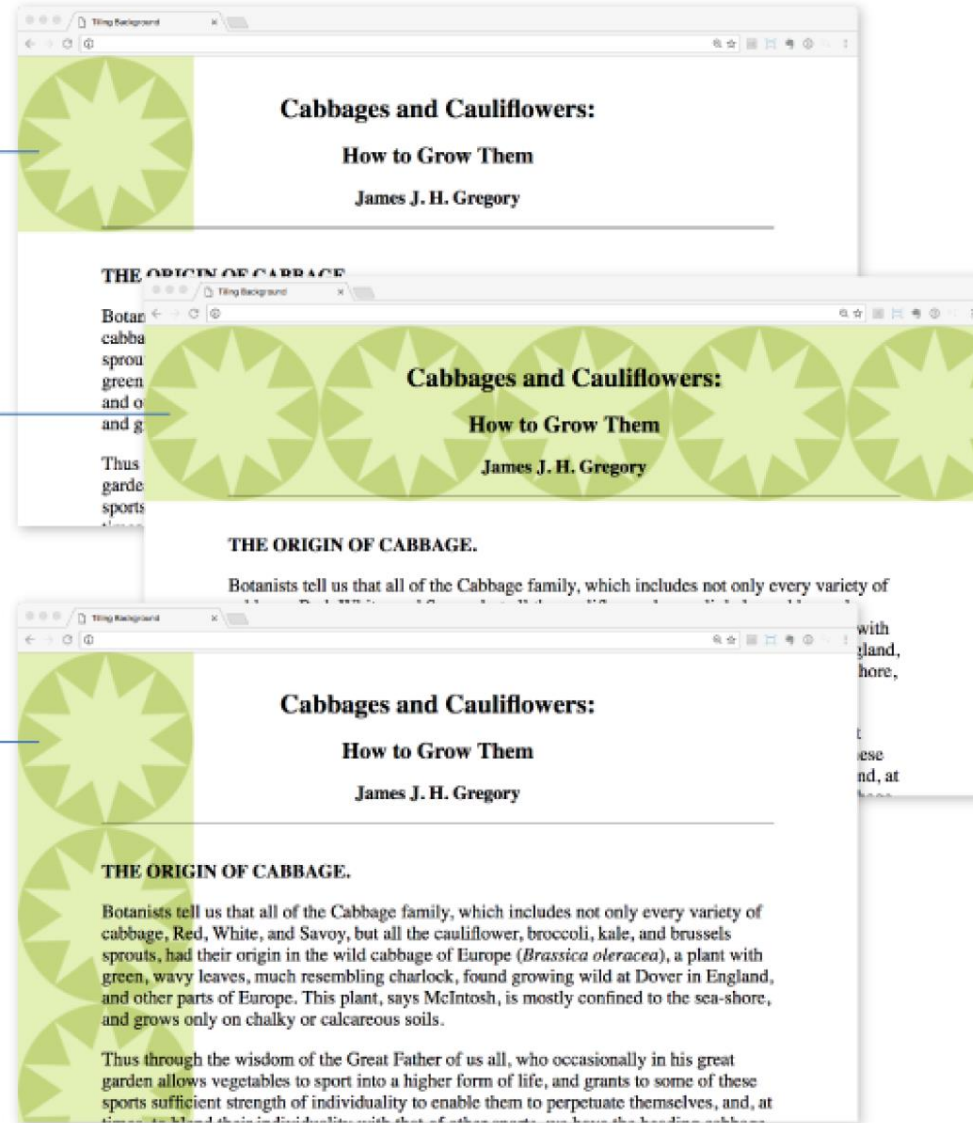
Background-Repeat

I Sami Rakha

no-repeat

repeat-x

repeat-y



Background-Position

- **Values:** Length, percentage, left, center, right, top, bottom
- Specifies the position of the **origin image**, the first image that is placed in the background from which tiling images extend.

Examples (horizontal position goes first):

```
background-position: left bottom;
```

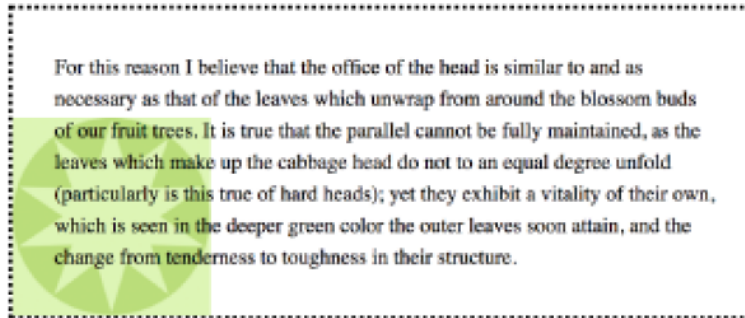
```
background-position: 300px 100px;
```

```
background-position: 25% 100%;
```

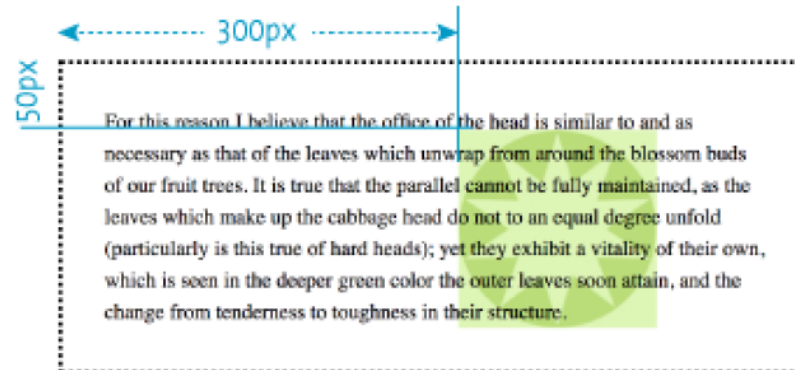
- value consists of two tokens, each of which can be top, left, right, bottom, center, a percentage, or a length value in px, pt, etc.
- value can be negative to shift left/up by a given amount

Background-Position

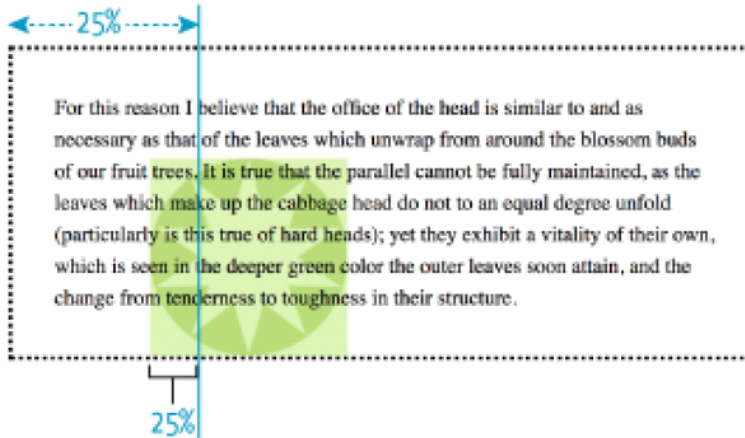
Rakha



`background-position: left bottom;`



`background-position: 300px 50px;`



`background-position: 25% 100%;`

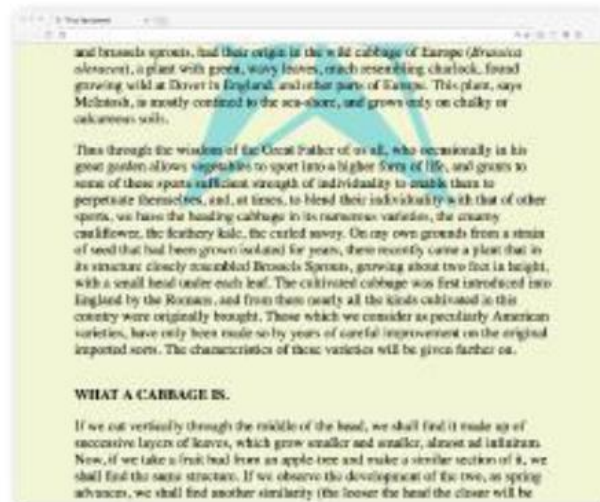
Background Attachment

- **Values:** scroll, fixed, local
- Specifies whether the background image scrolls with the content or stays in a fixed position relative to the viewport.

Dr. Mohamed Sami Rakha



A large non-repeating background image in the body of the document.



background-attachment: **scroll**;

By default, the background image is attached to the body element and scrolls off the page when the page content scrolls.



background-attachment: **fixed**;

When background-attachment is set to fixed, the image stays in its position relative to the browser viewing area and does not scroll with the content.

Background Size

- **Values:** Length, percentage, auto, cover, contain
- Specifies the size of the tiling image: controls **how a background image is scaled or stretched** to fit its element.



target.png
300 × 300 pixels

WHAT A CABBAGE IS.

If we cut vertically through the middle of the head, we shall find it made up of successive layers of leaves, which grow smaller and smaller, almost ad infinitum. Now, if we take a fruit bud from an apple-tree and make a similar section of it, we shall find the same structure. If we observe the development of the two, as spring advances, we shall find another similarity (the looser the head the closer will be the resemblance),—the outer leaves of each will unwrap and unfold, and a flower stem will push out from each. Here we see that a cabbage is a bud, a seed bud (as all fruit buds may be termed, the production of seed being the primary object in nature, the fruit enclosing it playing but a secondary part), the office of the leaves being to cover, protect, and afterwards nourish the young seed shoot.

The outer leaves which surround the head appear to have the same office as the leaves which surround the growing fruit bud, and that office closes with the first year, as does that of the leaves surrounding fruit buds, when each die and drop off. In my locality the public must have perceived more or less clearly the analogy between the heads of cabbage and the buds of trees, for when they speak of small heads they frequently call them "buds." That the close wrapped leaves which make the cabbage head and surround the seed germ, situated just in the middle of the head at the termination of the stump, are necessary for its protection and nutrition when young, is proved, I think, by the fact that those cabbages, the heads of which are much decayed, when set out for seed, no matter how sound the seed germ may be at the end of the stump, never make so large or healthy a seed shoot as those do the heads of which are sound; as a rule, after pushing a feeble growth, they die.

background-size: 600px 300px;

WHAT A CABBAGE IS.

If we cut vertically through the middle of the head, we shall find it made up of successive layers of leaves, which grow smaller and smaller, almost ad infinitum. Now, if we take a fruit bud from an apple-tree and make a similar section of it, we shall find the same structure. If we observe the development of the two, as spring advances, we shall find another similarity (the looser the head the closer will be the resemblance),—the outer leaves of each will unwrap and unfold, and a flower stem will push out from each. Here we see that a cabbage is a bud, a seed bud (as all fruit buds may be termed, the production of seed being the primary object in nature, the fruit enclosing it playing but a secondary part), the office of the leaves being to cover, protect, and afterwards nourish the young seed shoot.

The outer leaves which surround the head appear to have the same office as the leaves which surround the growing fruit bud, and that office closes with the first year, as does that of the leaves surrounding fruit buds, when each die and drop off. In my locality the public must have perceived more or less clearly the analogy between the heads of cabbage and the buds of trees, for when they speak of small heads they frequently call them "buds." That the close wrapped leaves which make the cabbage head and surround the seed germ, situated just in the middle of the head at the termination of the stump, are necessary for its protection and nutrition when young, is proved, I think, by the fact that those cabbages, the heads of which are much decayed, when set out for seed, no matter how sound the seed germ may be at the end of the stump, never make so large or healthy a seed shoot as those do the heads of which are sound; as a rule, after pushing a feeble growth, they die.

background-size: cover;

The entire background area of the element is covered, and the image maintains its proportions even if it is clipped.

WHAT A CABBAGE IS.

If we cut vertically through the middle of the head, we shall find it made up of successive layers of leaves, which grow smaller and smaller, almost ad infinitum. Now, if we take a fruit bud from an apple-tree and make a similar section of it, we shall find the same structure. If we observe the development of the two, as spring advances, we shall find another similarity (the looser the head the closer will be the resemblance),—the outer leaves of each will unwrap and unfold, and a flower stem will push out from each. Here we see that a cabbage is a bud, a seed bud (as all fruit buds may be termed, the production of seed being the primary object in nature, the fruit enclosing it playing but a secondary part), the office of the leaves being to cover, protect, and afterwards nourish the young seed shoot.

The outer leaves which surround the head appear to have the same office as the leaves which surround the growing fruit bud, and that office closes with the first year, as does that of the leaves surrounding fruit buds, when each die and drop off. In my locality the public must have perceived more or less clearly the analogy between the heads of cabbage and the buds of trees, for when they speak of small heads they frequently call them "buds." That the close wrapped leaves which make the cabbage head and surround the seed germ, situated just in the middle of the head at the termination of the stump, are necessary for its protection and nutrition when young, is proved, I think, by the fact that those cabbages, the heads of which are much decayed, when set out for seed, no matter how sound the seed germ may be at the end of the stump, never make so large or healthy a seed shoot as those do the heads of which are sound; as a rule, after pushing a feeble growth, they die.

background-size: contain;

The image is sized proportionally so it fits entirely in the element. There may be room left over for tiling (as shown).

Gradient Fills

A **gradient** is a transition from one color to another.

Linear gradients change colors along a line.

Radial gradients start at a point and spread outward in a circular or elliptical shape.

You can generate a gradient image for use as a background using **linear-gradient()** and **radial-gradient()** notation.

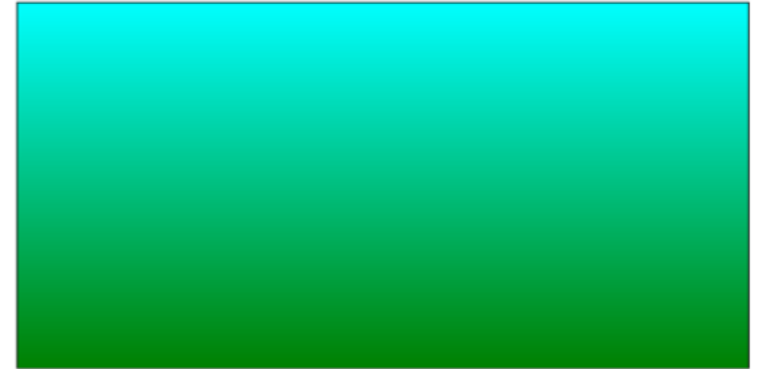
Example:

```
#banner {  
  background-image: linear-gradient(180deg, aqua, green);  
}
```

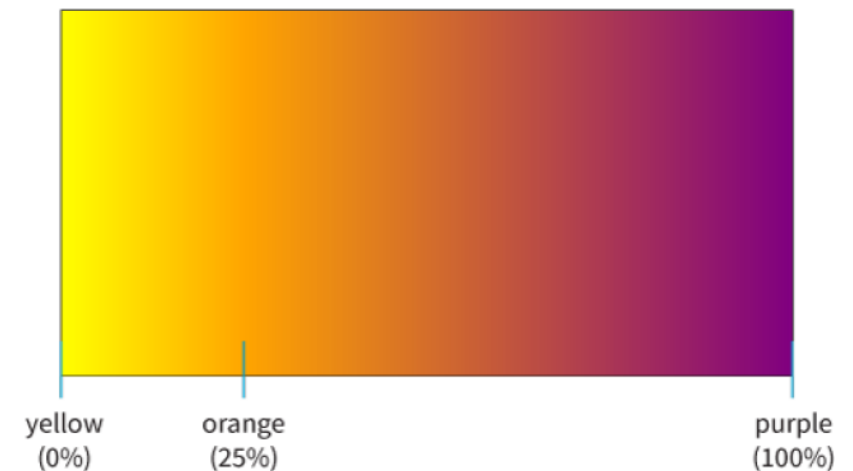
Linear Gradient

- The **linear-gradient()** notation provides the angle of a gradient line and the colors the line passes through.
- It is specified in degrees (deg) or keywords (to top, to right, to bottom, to left).

```
linear-gradient(180deg, aqua, green);  
or  
linear-gradient(to bottom, aqua, green);
```



```
linear-gradient(90deg, yellow, orange 25%, purple);
```



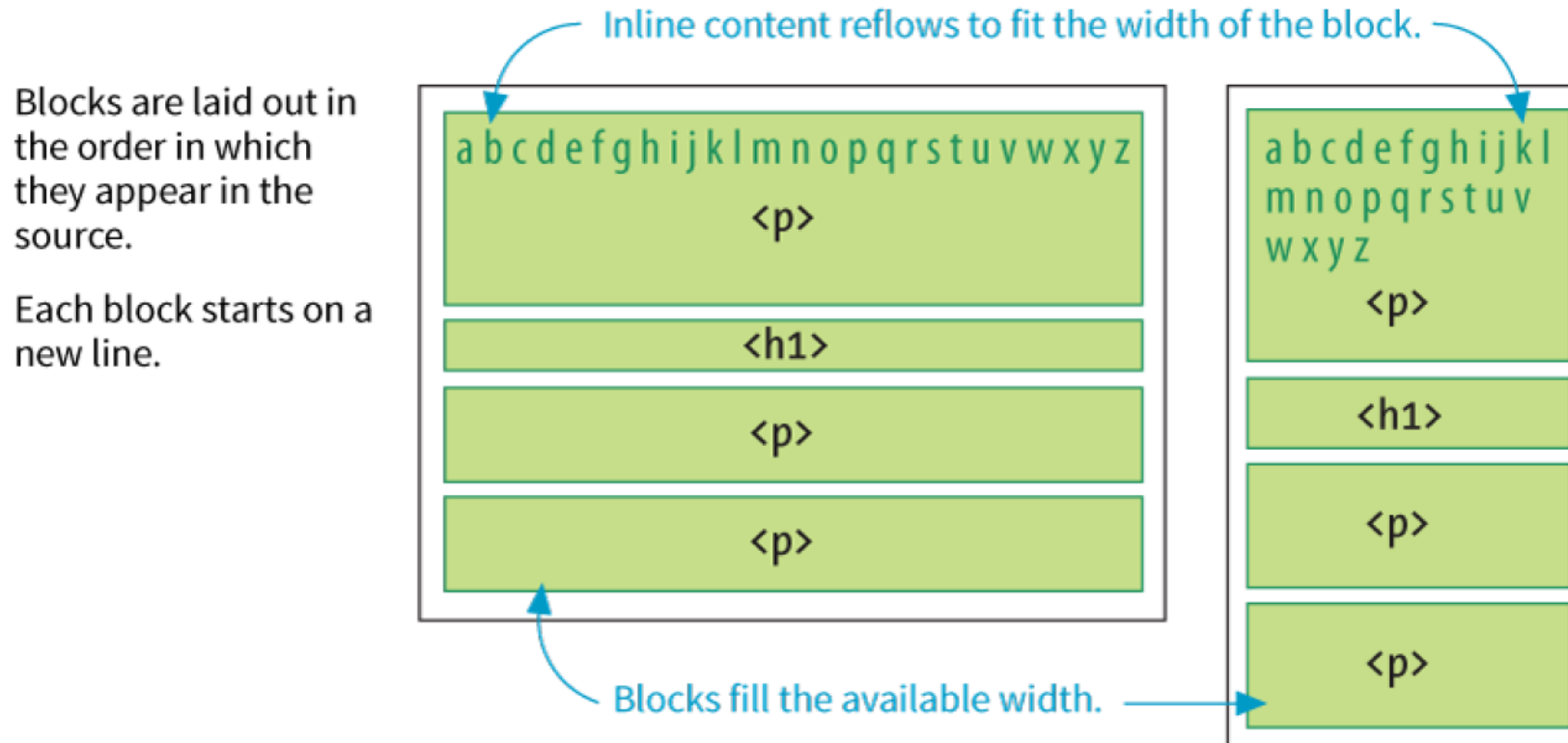
CSS Properties for Backgrounds

Property	Description
background-color	color to fill background
background-image	image to place in background
background-position	placement of bg image within element
background-repeat	whether/how bg image should be repeated
background-attachment	whether bg image scrolls with page
background	shorthand to set all background properties

Floating & Positioning

Normal Flow

In the normal flow , elements are laid out from top to bottom in the order in which they appear in the source and from left to right (in left to right reading languages).



Floating

The **float property** is used to move (or “float”) an element to the left or right side of its container, allowing text or other inline content to wrap around it.

Inline image floated to the right

```
img { float: right; }
```

Image moves over, and text wraps around it

After the cream is frozen rather stiff, prepare a tub or bucket of coarsely chopped ice, with one-half less salt than you use for freezing. To each ten pounds of ice allow one quart of rock salt. Sprinkle a little rock salt in the bottom of your bucket or tub, then put over a layer of cracked ice, another layer of salt and cracked ice, and on this stand your mold, which is not filled, but is covered with a lid, and pack it all around, leaving the top, of course, to pack later on. Take your freezer near this tub. Remove the lid from the mold, and pack in the cream, smoothing it down until you have filled it to overflowing. Smooth the top with a spatula or limber knife, put over a sheet of waxed paper and adjust the lid.



Floating

- Floated elements are removed from the normal flow but influence the surrounding content.
- Floated elements stay within the content area of the element that contains it.
- Margins are always maintained (they don't collapse) on all sides of floated elements.
- You must provide a width for a floated text element (because the default width is auto).
- **Elements don't float higher than their reference in the source.**

Clearing Floated Elements

Clear , Values: left , right , both , none

Dr. Mohamed Sami Rakha

Prevents an element from appearing next to a floated element and forces it to start against the next available “clear” space

```
img {  
  float: left;  
  margin-right: .5em;  
}  
h2 {  
  clear: left;  
  margin-top: 2em;  
}
```



If pure raw cream is stirred rapidly, it swells and becomes frothy, like the beaten whites of eggs, and is "whipped cream." To prevent this in making Philadelphia Ice Cream, one-half the cream is scalded, and when it is very cold, the remaining half of raw cream is added. This gives the smooth, light and rich consistency which makes these creams so different from others.

USE OF FRUITS

Use fresh fruits in the summer and the best canned unsweetened fruits in the winter. If sweetened fruits must be used, cut down the given quantity of sugar. Where acid fruits are used, they should be added to the cream after it is partly frozen.

The time for freezing varies according to the quality of cream or milk or water; water ices require a longer time than ice creams. It is not well to freeze the mixtures too rapidly; they are apt to be coarse, not smooth, and if they are churned before the mixture is icy cold they will be greasy or "buttery."

(The h2 is “cleared” and starts below the floated element.)

Positioning

Using CSS, you can place elements exactly on a page using a technique called “**positioning**.” Positioning is determined by an **X-axis** and a **Y-axis**. To specify a specific point on the screen, you can use the **X** and **Y** coordinates for that point.

There are several ways to specify position in CSS: ***absolute, relative, fixed, inherit, and static.***

The three most often used are *absolute, relative, and fixed.*

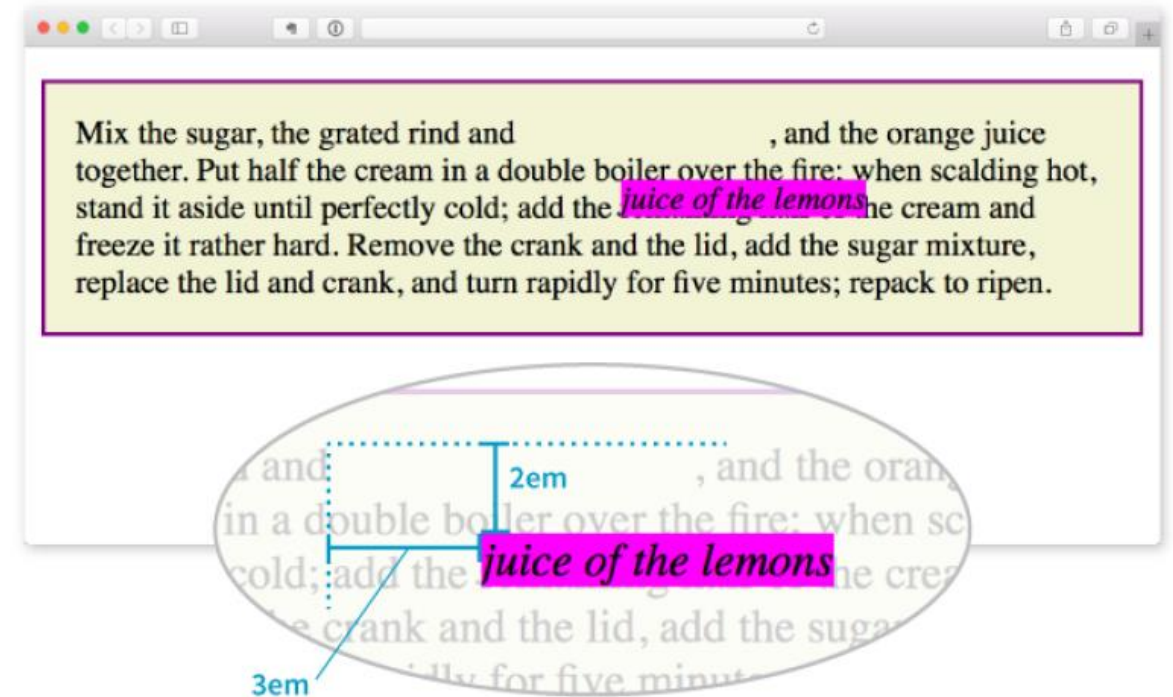
Positioning

Position Type	Behavior	Example Use Case
Static (default)	Follows normal document flow	Default paragraphs, divs
Relative	Moves relative to its normal position, still takes up space	Fine-tuning layout offsets
Absolute	Removed from normal flow; positioned relative to nearest ancestor with positioning	Tooltips, pop-ups inside a container
Fixed	Positioned relative to viewport; stays in place when scrolling	Sticky navigation bar, floating chat button
Inherit	Takes the position value of the parent element	Maintaining consistent positioning in nested elements

Relative Positioning

- Moves the element relative to its original position
- The space it originally occupied is preserved.

```
em {  
  position: relative;  
  top: 2em; /* moves it down */  
  left: 3em; /* moves it right */  
  background-color: fuchsia;  
}
```

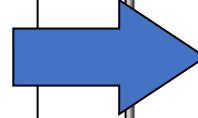


Positioning – Relative Example

Dr. Mohamed Sami Rakha

```
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>Offset example</title>
<style type="text/css">
<!--
.offset {
  position: relative;
  top: 150px;
  left: 50px;
}
-->
</style>
</head>

<body>
<p>This text will flow normally across
the screen. The next line will be offset
from the end of this line.
<span class="offset">This text is offset
from the line above. Neat, huh?</span>
</p>
</body>
</html>
```



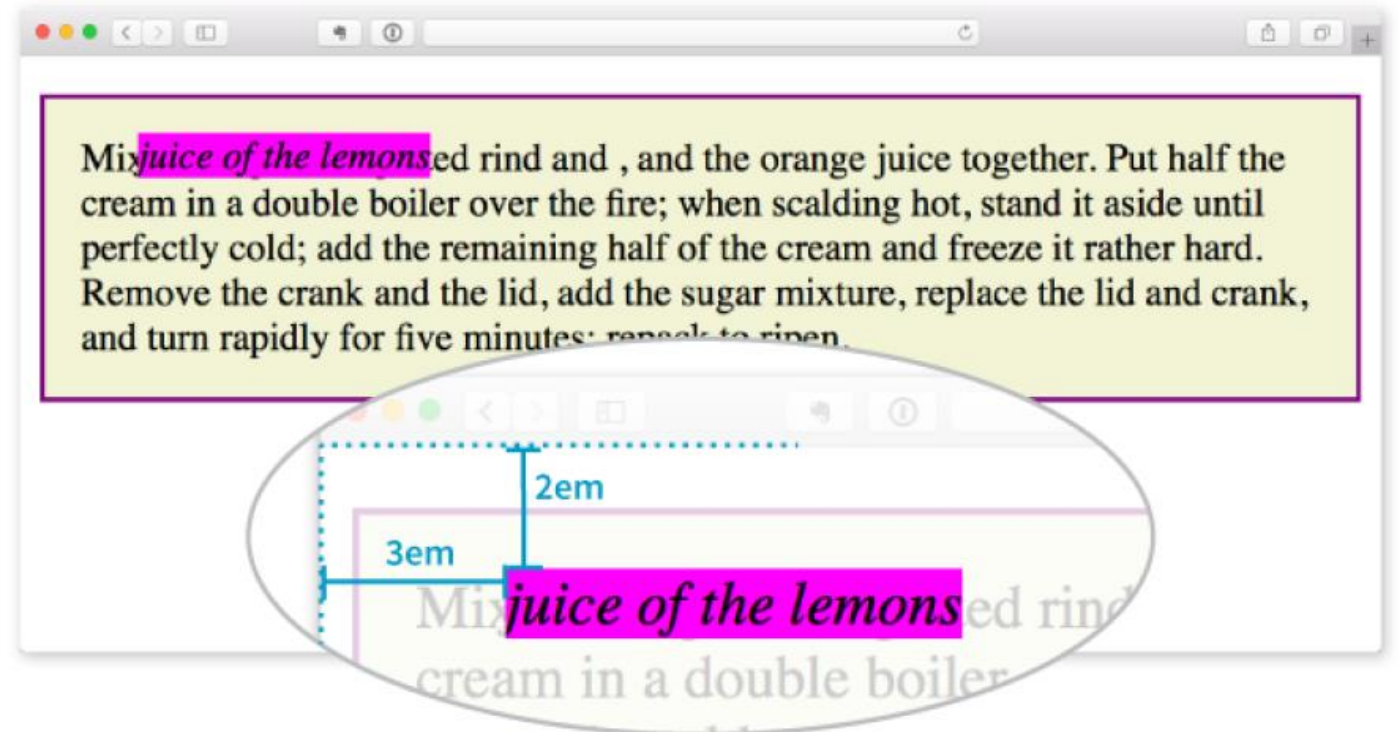
This text will flow normally across the screen. The next line will be offset from the end of this line.

This text is offset from the line above. Neat, huh?

Positioning – Absolute Example

- Moves the element relative to the viewport or containing block element
- The space it originally occupied is closed up.

```
em {  
  position: absolute;  
  top: 2em;  
  left: 3em;  
  background-color: fuchsia;  
}
```

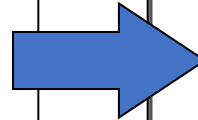


Positioning – Absolute Example

Dr. Mohamed Sami Rakha

```
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>Pink Box</title>
<style type="text/css">
<!--
#pinkbox {
    position: absolute;
    width: 200px;
    background-color: pink;
    left: 0px;
    top: 150px;
    height: 200px;
}
-->
</style>
</head>

<body>
<div id="pinkbox">This is a 200
by 200 pixel pink box located
150 pixels from the top of the
web page.
</div>
</body>
</html>
```

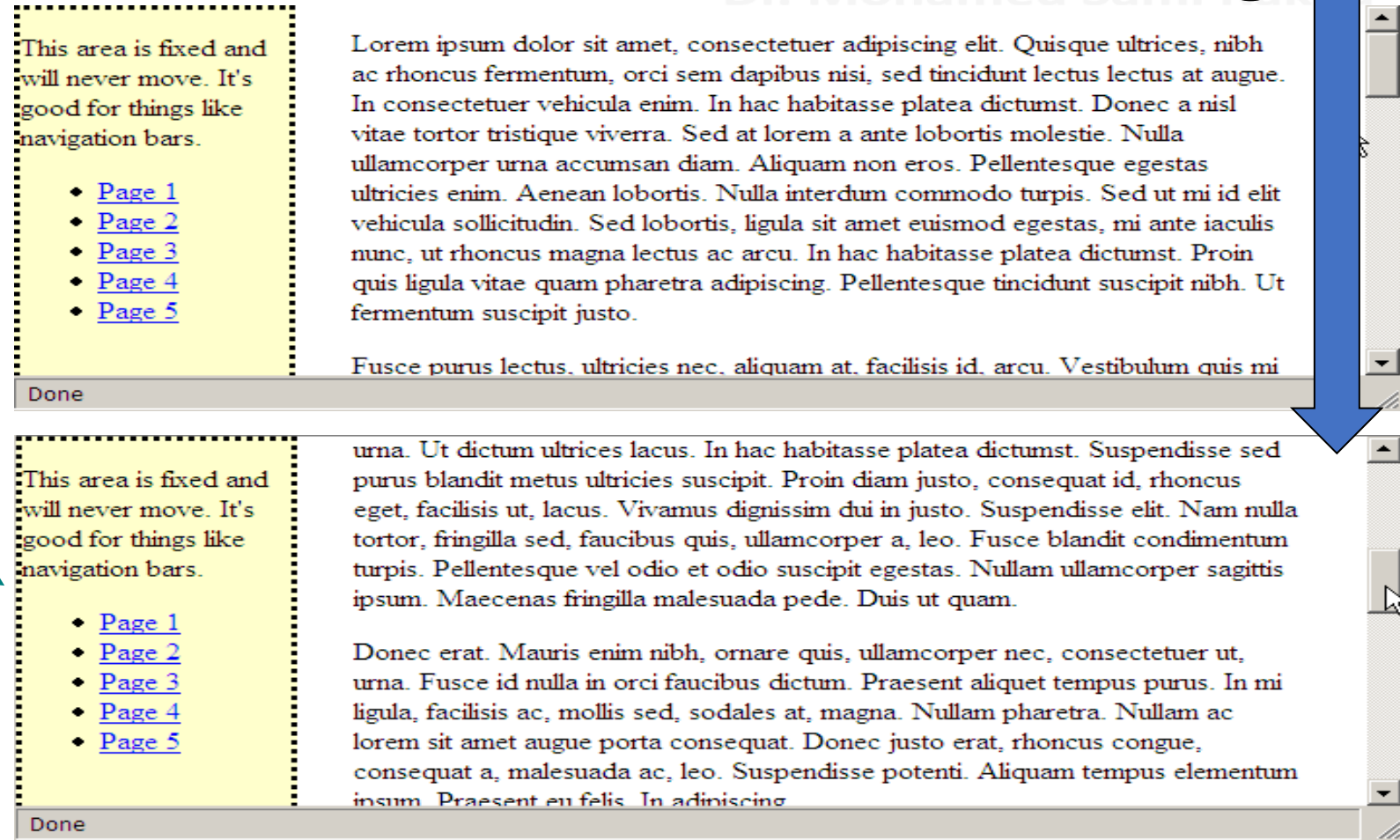


This is a 200 by 200 pixel pink box located 150 pixels from the top of the web page.

Positioning – Fixed Example

Scroll Down with Fixed Positioning

Positioned relative to viewport; stays in place when scrolling

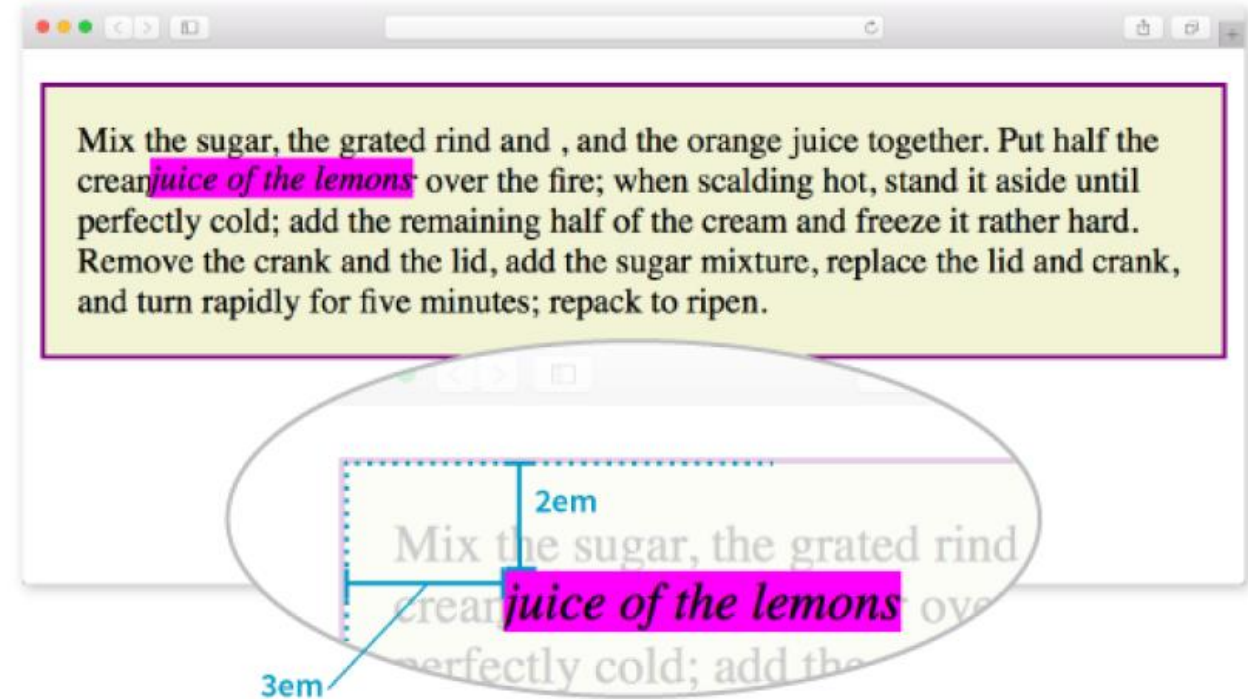


Containing Blocks Rules

- A positioned element serves as a **containing block** (or positioning context) for the elements it contains.
- If a positioned element has an ancestor that has its **position** set to relative, absolute, or fixed, then its position will be relative to that containing block element.
- If a positioned element is not contained within another positioned element, then it is placed relative to the initial containing block (the **HTML** element) and the viewport.

Containing Blocks Rules

```
p {  
  position: relative;  
  padding: 15px;  
  background-color: #F2F5D5;  
  border: 2px solid purple;  
}  
em {  
  position: absolute;  
  top: 2em;  
  left: 3em;  
  background-color: fuchsia;  
}
```



The relatively positioned `p` element acts as a containing block for the `em` element.

Dr. Mohamed Sami Rakha

Thank You...